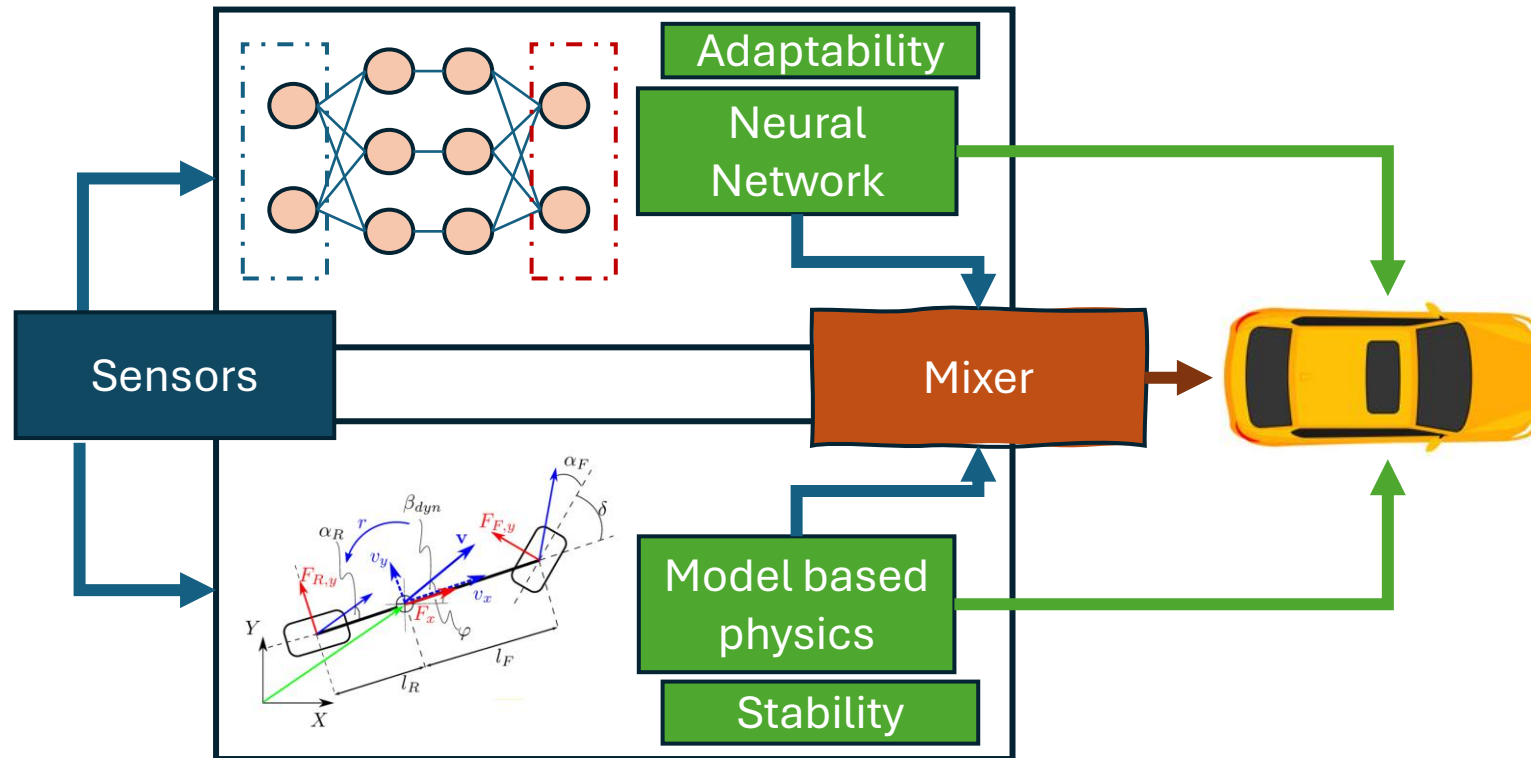


Robust KalmanNet State Estimation



Problem Setup: Signals and Measurement

Measurement vector

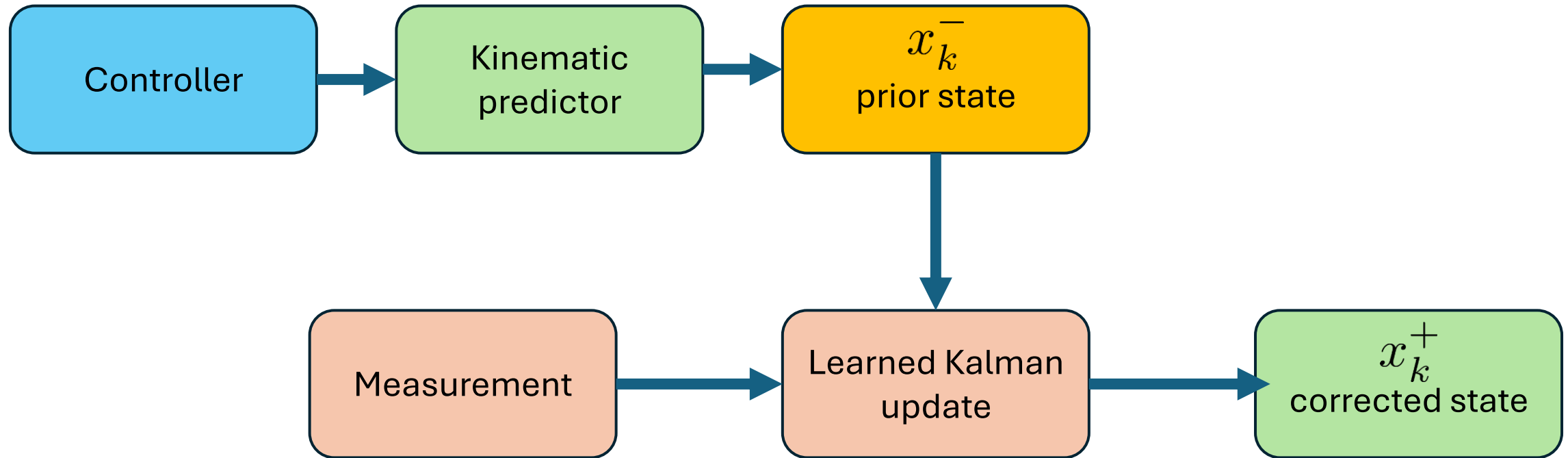
$$z_k = \begin{bmatrix} x_{gps,k} & y_{gps,k} & \psi_k & v_k \end{bmatrix}^T$$

When **attack in the sensor layer** , its will be affected the performance of the state estimator .

Break down all the chain of estimation before

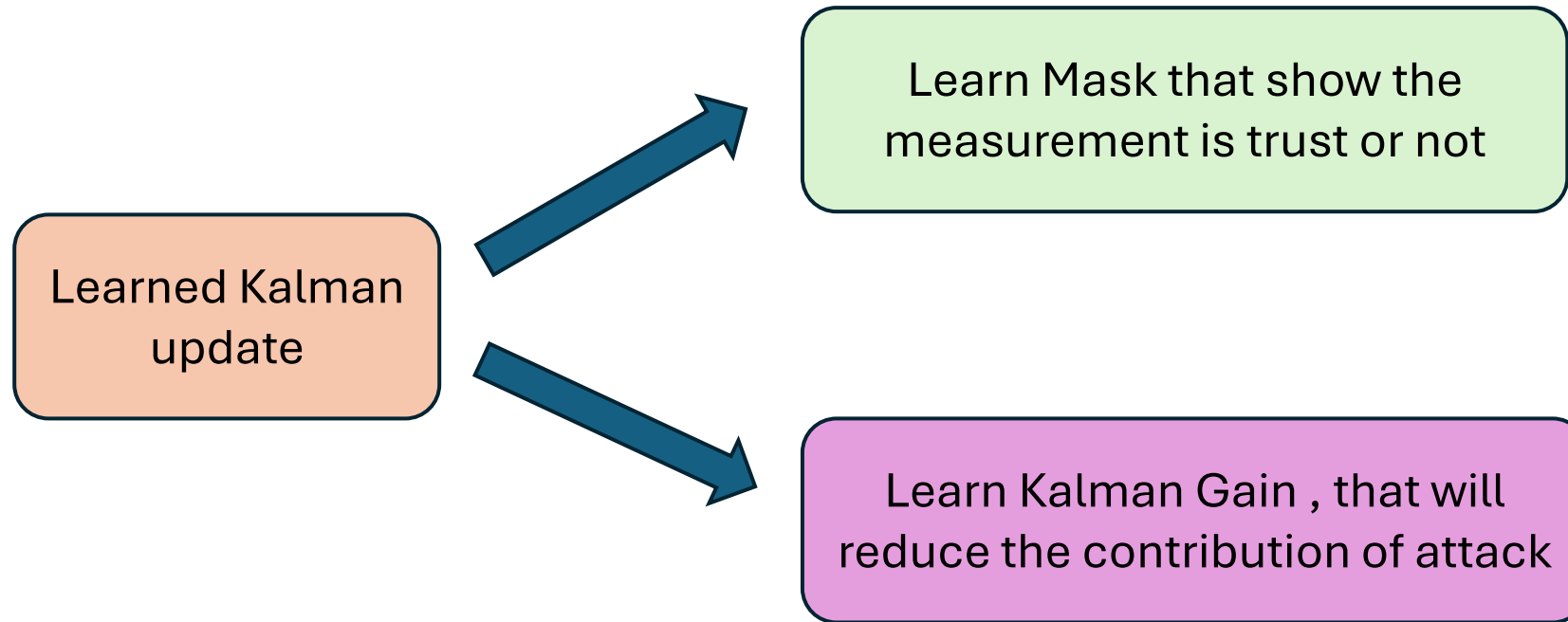
RKNet keep the performance while rejecting corrupted GPS and velocity measurements.

RKNet Concept Design



- Prediction is physics-based.
- Update learns measurement reliability mask and Kalman gain.
- Main robustness target: GPS position and velocity corruption.

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Prediction Math: Bicycle Model

$$x_k^- = x_{k-1}^+ + v_k \cos(\psi_{k-1}^+) \Delta t$$

$$y_k^- = y_{k-1}^+ + v_k \sin(\psi_{k-1}^+) \Delta t$$

$$\psi_k^- = \psi_{k-1}^+ + \frac{v_k \tan(\delta_k)}{L} \Delta t$$

$$v_k^- = v_{k-1}^+ + a_k \Delta t$$

$$a_k = \frac{-v_{k-1}^+ + u_k}{\tau}$$

Config values: $L = 0.2$, $\tau = 0.301$, $g = 6.598$, $|v| \leq 2.0$.

Learn Robust Update Math

1. Innovation

$$\mathbf{r}_k = \mathbf{z}_k - H\mathbf{x}_k^-$$

2. Prediction difference

$$\Delta\mathbf{x}_k = \mathbf{x}_k^- - \mathbf{x}_{k-1}^+$$

3. Measurement difference

$$\Delta\mathbf{z}_k = \mathbf{z}_k - \mathbf{z}_{k-1}$$

4. Measurement availability

$$a_k = [\textit{gps}_{valid}, \textit{gps}_{valid}, \theta_{valid}, 1]$$

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Updater features

$$\mathbf{f}_k = [\Delta\mathbf{x}_k, \mathbf{r}_k, \Delta\mathbf{z}_k, \mathbf{a}_k]$$

Measurement mask



$$\mathbf{m}_k^z = \sigma(\textit{MLP}(\mathbf{f}_k))$$

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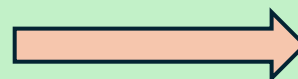
Measurement mask

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Learned Kalman gain

$$\mathbf{h}_k = \textit{GRU}(\mathbf{m}_k^z \odot \mathbf{f}_k, \mathbf{h}_{k-1})$$



$$\mathbf{K}_k = \tanh(\textit{MLP}(\mathbf{h}_k))$$

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Final update

$$\mathbf{x}_k^+ = \mathbf{x}_k^- + K_k \tilde{\mathbf{r}}_k \quad \text{where} \quad \tilde{\mathbf{r}}_k = \mathbf{m}_k^z \odot \mathbf{r}_k$$

Attack Setup: Focus on GPS and Velocity

GPS position attacks

- Noise: $z_{xy}^{\text{atk}} = z_{xy} + \epsilon$
- Jump: $z_{xy}^{\text{atk}} = z_{xy} + \mathbf{b}_{\text{jump}}$
- Freeze: $z_{xy,k}^{\text{atk}} = z_{xy,k_0}, \quad k \geq k_0$
- Dropout: $\text{gps_valid}_k = 0$

Velocity attacks

- Bias: $z_v^{\text{atk}} = z_v + b_v$
- Scale: $z_v^{\text{atk}} = \alpha z_v$
- Freeze: $z_{v,k}^{\text{atk}} = z_{v,k_0}, \quad k \geq k_0$
- Noise: $z_v^{\text{atk}} = z_v + \epsilon_v$
- Zero-out: $z_{v,k}^{\text{atk}} = 0$

Attack labels supervise the mask:

$$\text{label}_{k,i} = \begin{cases} 1, & \text{measurement channel } i \text{ attacked} \\ 0, & \text{clean} \end{cases}$$

$$m_{k,i}^{z,\text{target}} = 1 - \text{label}_{k,i}$$

$$m_{k,i}^{z,\text{target}} = \begin{cases} 1, & \text{trust measurement channel } i \\ 0, & \text{suppress measurement channel} \end{cases}$$

Saved fields: raw sensors, GPS flags, z_k , target $\mathbf{x}_k^*$, timestamps.

Training schedule

- Phase A : Train updater only. Train updater with good previous state input
- Phase B : End-to-end fine tuning. Model uses its own previous output train the full recurrent estimation loop
- Validation split: 20% tail.

Loss

$$\mathcal{L} = \lambda_u L_{upd} + \lambda_p L_{pred} + \lambda_m L_{mask} + \lambda_K L_{gain} + \lambda_{\Delta K} L_{\Delta K} + \lambda_{\Delta m} L_{\Delta m}$$

$$L_{upd} = MSE_W(\mathbf{x}^+, \mathbf{x}^*), \quad L_{pred} = MSE_W(\mathbf{x}^-, \mathbf{x}^*)$$

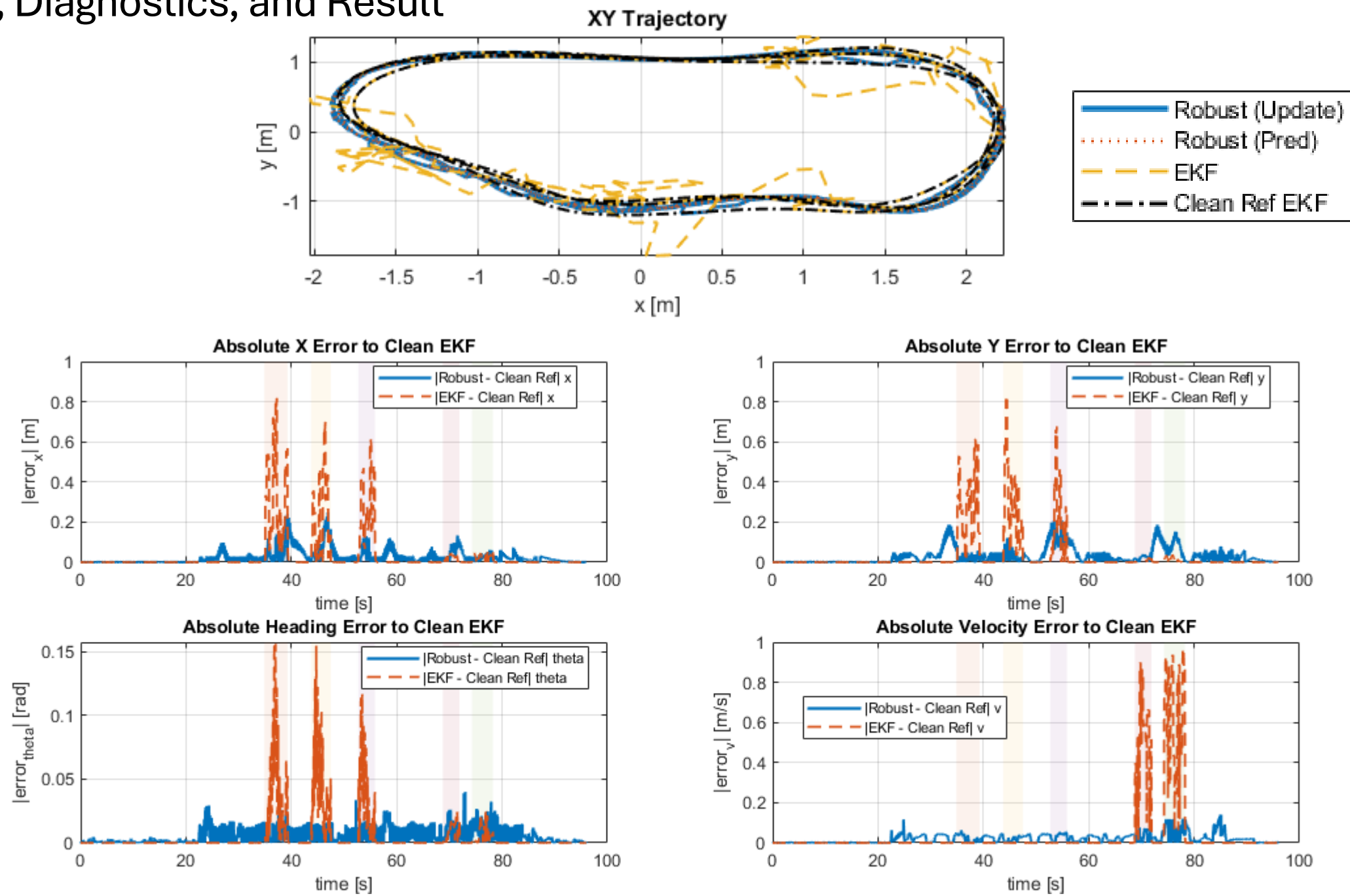
$$L_{mask} = BCE(\mathbf{m}^z, \mathbf{m}^{z,target}), \quad L_{gain} = \|K\|_2^2$$

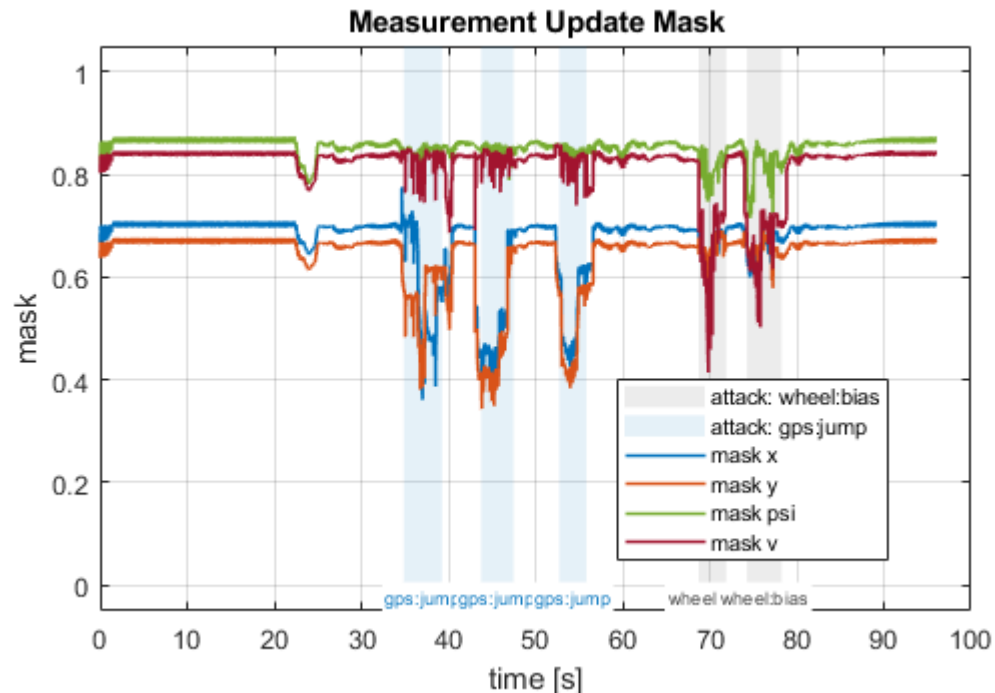
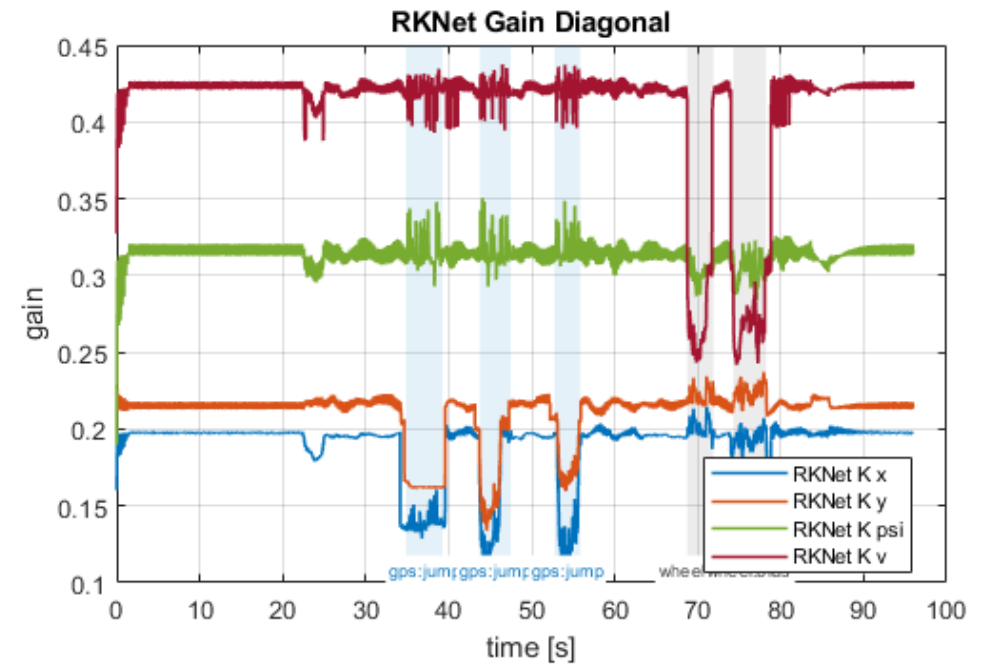
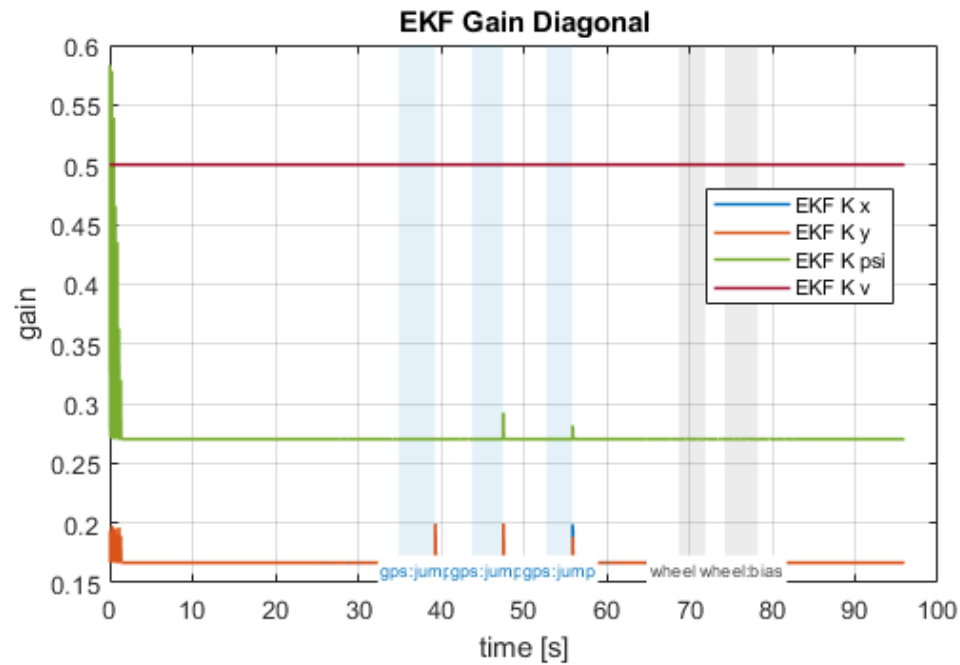
$$L_{\Delta K} = \|K_k - K_{k-1}\|_2^2, \quad L_{\Delta m} = \|\mathbf{m}_k^z - \mathbf{m}_{k-1}^z\|_2^2$$

- L_{upd} : final corrected state should match the reference.
- L_{pred} : prior prediction should stay physically reasonable.
- L_{mask} : clean channel target = 1, attacked channel target = 0.
- L_{gain} : avoid very large Kalman gains.
- $L_{\Delta K}$: avoid gain chatter.
- $L_{\Delta m}$: avoid mask flicker.

$$\lambda_u = 0.85, \lambda_p = 0.0, W = [1.0, 1.0, 2.5, 1.2], \lambda_K = 0.01, \lambda_{\Delta K} = 0.02.$$

Testing, Diagnostics, and Result





Mask and Gain RKNet go down ↓

So less depend on the measurement ,
more robust